



Name: \_\_\_\_\_ Cohort/Batch: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_

## JAVASCRIPT PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Programming Languages

TOTAL: 10 QUESTIONS

**Q1** What are the primary design goals of JavaScript?

Threat Model

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**Q6** How does JavaScript implement object-oriented or functional programming?

Observability

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**Q2** How does JavaScript handle type checking: static or dynamic?

Architecture

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**Q7** How does JavaScript handle errors?

Lifecycle

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**Q3** How does JavaScript manage memory?

Configuration

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**Q8** What testing tools are commonly used in JavaScript?

Compliance

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**Q4** What is JavaScript's concurrency model?

Hardening

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**Q9** What makes JavaScript well-suited for its target domain?

Testing

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**Q5** How are packages and dependencies managed in JavaScript?

SSO / IdP

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**Q10** What are common performance pitfalls in JavaScript?

Tuning

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ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 JavaScript was designed with specific priorities

Mitigation

JavaScript was designed with specific priorities around performance, safety, or developer productivity depending on its domain. These goals shape its syntax, type system, and standard library choices.

A6 JavaScript supports classes and inheritance

Observability

JavaScript supports classes and inheritance for OOP, or first-class functions and immutability for FP, or both. Many modern JavaScript programs blend both paradigms depending on the problem.

A2 JavaScript uses its type system to

Primitives

JavaScript uses its type system to catch errors at compile time (static) or at runtime (dynamic). This affects refactoring safety, tooling support, and the kinds of bugs that reach production.

A7 JavaScript surfaces errors via exceptions,

Automation

JavaScript surfaces errors via exceptions, Result/Either types, or error return values. The choice impacts whether errors are checked at compile time and how calling code is structured.

A3 JavaScript uses one of three approaches:

Hardening

JavaScript uses one of three approaches: manual allocation/deallocation, a garbage collector that periodically reclaims unreachable objects, or automatic reference counting. Each has different latency and throughput tradeoffs.

A8 JavaScript has a standard or widely

Governance

JavaScript has a standard or widely adopted test runner and assertion library. Tests are typically organized into unit, integration, and end-to-end layers with coverage reporting built in or via plugins.

A4 JavaScript supports concurrency through threads,

Gotchas

JavaScript supports concurrency through threads, coroutines, async/await, or an actor model. The choice determines how I/O-bound and CPU-bound workloads are scaled and how shared state is safely accessed.

A9 JavaScript excels in its domain due

Assessment

JavaScript excels in its domain due to a combination of performance characteristics, ecosystem maturity, language ergonomics, and community support that makes common tasks simple.

A5 JavaScript uses a package manager and

Federation

JavaScript uses a package manager and a manifest file to declare dependencies and version constraints. A lock file pins exact versions to ensure reproducible builds across environments.

A10 Typical performance issues in JavaScript

Optimization

Typical performance issues in JavaScript include excessive allocations, blocking the main thread or event loop,  $O(n^2)$  data structures, and missing caching. Profiling and benchmarking tools are available to diagnose them.